WORKSHEET

Expected values

1 A 4-sided die is rolled and the results recorded as follows: 3, 1, 1, 4, 3, 2, 2, 3, 1, 2, 1, 4, 1, 3, 3, 2.

Random variable X = the value displayed on the die.

a Use these results to estimate the probability distribution for X.

b What is the average number showing on the die?

2 Students have been asked how many sports they play. Results are as follows:

2, 0, 3, 1, 0, 1, 2, 4, 1, 3, 0, 2, 2, 4, 1, 1, 2, 1, 3, 0, 1, 0, 0, 2, 2, 1, 2, 1, 0, 2

a Find the probability distribution of S, the number of sports played.

b What is the average number of sports played by the students?

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3 A sample of the number of mobile phones per household was recorded as follows:

Number of phones	Frequency
0	1
1	1
2	3
3	7
4	11
5	8
6	4

a What is the probability that a house will have 5 phones?

b What is the probability that a house will have fewer than 3 phones?

c Use the data to find the average number of phones per household.

4 The number of runs scored per over by a team playing 20-20 cricket was as follows:

9, 7, 10, 16, 10, 6, 5, 4, 6, 2, 5, 5, 6, 9, 7, 8, 9, 11, 20, 14

What is the run rate required (runs per over) by the opposing team to win the match?

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5 A deck of cards has had the jokers and aces removed (jacks, queens and kings are picture cards). If picture cards have a value of ten, find:

a the average value of the 48 cards

b the probability distribution for X, the value of a randomly chosen card

c the probability that a randomly drawn card will have a value of 10

d the probability that a randomly drawn card will not have a value of 10.

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6 Two standard six-sided dice are rolled and random variable X = the sum of the two numbers showing on the dice.

a Determine the probability distribution of X.

b What is the average value of X?

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7 Two coins are tossed and random variables X and Y are defined as follows:

Outcome	X	Y
TT	0	0
TH	1	1
HT	1	2
HH	2	3

a Determine the probability distribution of X.

b What is the average value of X?

c Determine the probability distribution of Y.

d What is the average value of Y?

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Answers

1 a

Outcome (x)	Frequency	$P(X = x)$
1	5	0.3125
2	4	0.25
3	5	0.3125
4	2	0.125
Total	16	1

b 2.25

2 a

Number of sports (s)	Frequency	$P(S = s)$
0	7	0.2333
1	9	0.3
2	9	0.3
3	3	0.1
4	2	0.0667
Total	30	1

b ≈ 1.4667

3 a ≈ 0.2286

b ≈ 0.1429

c ≈ 3.89

4 8.5 runs per over

5 a 7

b

Card value (x)	Frequency	$P(X = x)$
2	4	0.0833
3	4	0.0833
4	4	0.0833
5	4	0.0833
6	4	0.0833
7	4	0.0833
8	4	0.0833
9	4	0.0833
10	16	0.3333
Total	48	1

c ≈ 0.3333

d ≈ 0.6667

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6 a

Outcome (x)	Frequency	P($X = x$)
2	1	0.0278
3	2	0.0556
4	3	0.0833
5	4	0.1111
6	5	0.1389
7	6	0.1667
8	5	0.1389
9	4	0.1111
10	3	0.0833
11	2	0.0556
12	1	0.0278
Total	36	1

b 7

7 a

x	Frequency	$P(X = x)$
0	1	0.25
1	2	0.5
2	1	0.25
Total	4	1

b 1

c

y	Frequency	$P(Y = y)$
0	1	0.25
1	1	0.25
2	1	0.25
3	1	0.25
Total	4	1

d 1.5

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